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Wykonane pierwsze i drugie zadanie

```
clear all
```

```
clc
```

```
%% Zad1
```

```
A = 1;
```

```
t = 0:0.01:10;
```

```
p100 = (rand(1,100)-0.5)*2;
```

```
p1000 = (rand(1,1000)-0.5)*2;
```

```
p100(p100<0) = -A;
```

```
p100(p100>=0) = A;
```

```
p1000(p1000<0) = -A;
```

```
p1000(p1000>=0) = A;
```

```
%fazy = rand(1,6)*2*pi;
```

```
fazy = load("fazy.mat");
```

```
fazy = fazy.fazy;
```

```
psin = A*sin(pi*t*0.24 + fazy(1)) + A*sin(pi*t*0.42+ fazy(2)) + A*sin(pi*t*0.14+ fazy(3)) +  
A*sin(pi*t*0.74+ fazy(4)) + A*sin(pi*t*0.36+ fazy(5)) + A*sin(pi*t*0.66+ fazy(6));
```

```
plot(psin,'o')
```

```
%% Zad2
```

```
A = [1 -0.5 -0.7 0.5];
```

```
B = [0.7 0.3];
```

```
y100 = filter(B,A,p100);
```

```
plot(y100);
```

```
y1000 = filter(B,A,p1000);
```

```
plot(y1000)
```

```
ysin = filter(B,A,psin);
```

```
plot(ysin)
```

```
w100 = randn(size(y100)) * sqrt(0.1 * var(y100));
```

```
w1000 = randn(size(y1000)) * sqrt(0.1 * var(y1000));
```

```
wsin = randn(size(ysin)) * sqrt(0.1 * var(ysin));
```

```
%% Zad3
```

```
N = length(w100); % liczba pomiarów
```

```
M = 5; % liczba parametrów
```

```
for i=1:N-M
```

```
    for j=1:M+1
```

```
        U(i,j) = w100(N+2-j-i);
```

```
    end
```

```
end
```